ECE-006 Wi-Fi Hotspot Voucher & Session Accounting Engine

Design Document — Team E010

Voucher management: Greeshma Reddy R | Session management: Gowthami J | Language: Python 3.10+ (standard library only)

1. Overview

The engine is command-file driven (no UI). It reads **commands.txt**, loads voucher types and issued vouchers from CSV, processes a day of LOGIN / USAGE / LOGOUT events in timestamp order, and writes four files: **session_log.csv**, **voucher_report.csv**, **unused_vouchers.csv** and **errors.csv**.

Area	Owner	Files
Voucher management: types, issuing, activation, 24 h validity, balances, reports	Greeshma	models.py, voucher_manager.py, main.py, tests/test_vouchers.py
Session management: LOGIN/LOGOUT/USAGE, device limit, idle timeout, timestamp ordering, session log, rejection reasons	Gowthami	session_engine_corrected_v3.py

2. Data model

Class	Fields	Notes
VoucherType	prefix, voucher_type (TIME|DATA), quantity (minutes or MB), price, max_devices, type_id	type_id (VT-01 ...) is printed in the report
Voucher	code, voucher_type, issued_at, remaining_balance, activated, activation_time, expiry_time, active_devices, devices_used	Balance starts at the type's quantity. devices_used = every MAC that ever logged in
VoucherManager	voucher_types, types_by_id, vouchers, counters	Loads CSVs, looks vouchers up, produces the reports

3. Business-rule register

Every rule in the brief has an ID. *Owner* says which part of the code enforces it.

ID	Rule	Owner
BR-01	Activation: the 24 h window starts at the voucher's first successful LOGIN, not at issue. The window is [activation, activation + 24 h). A LOGIN at or after the end is rejected EXPIRED, including exactly 24 h.	Voucher
BR-02	Time vouchers: minutes consumed LOGIN to LOGOUT per device, summed across devices. A session that would exceed the balance is cut at the exact second the balance reaches zero. Balance never goes below 0 (charge_minutes).	Voucher + Session
BR-03	Data vouchers: MB from USAGE events are deducted. When remaining MB reaches zero, all sessions on that voucher end at that event's timestamp. Balance never goes below 0 (charge_mb).	Voucher + Session
BR-04	Concurrency: at most max_devices devices logged in at once; an extra LOGIN is rejected DEVICE_LIMIT.	Session
BR-05	Idle timeout: 10 minutes with no USAGE logs the device out at last activity + 10 min; time vouchers stop consuming from then.	Session
BR-06	Rounding: time is charged in whole minutes rounded up per session; data is charged per event rounded up to the next MB (math.ceil).	Voucher + Session
BR-07	Report: revenue counts a voucher on its day of activation; vouchers never activated are listed separately in unused_vouchers.csv.	Voucher
BR-08	Events are processed by timestamp, then LOGIN < USAGE < LOGOUT, then event_id, so out-of-order rows are handled.	Session
BR-09	Idle logouts and cut-offs due at or before an event's timestamp are applied before that event.	Session
BR-10	A LOGIN from a MAC already connected on that voucher is ignored: no new session, no error line.	Session
BR-11	LOGIN check order: unknown voucher, duplicate MAC, expired, balance exhausted, device limit.	Session + Voucher
BR-12	USAGE/LOGOUT with no active session (never logged in, or already idle-logged-out) is rejected NOT_LOGGED_IN and nothing is deducted.	Session

ID	Rule	Owner
BR-13	Malformed input rows are skipped and reported, never crash the run.	Both

4. Voucher module interface

The session engine uses only these calls; it never edits balances directly.

Call	Purpose	Rule
VoucherManager.load_types(path)	Read voucher_types.csv, validate and register each type	BR-13
VoucherManager.load_vouchers(path)	Read vouchers.csv; keep the given codes and issue times; skip unknown types and duplicate codes	BR-13
VoucherManager.get(code)	Return the voucher or None (None means UNKNOWN_VOUCHER)	BR-11
VoucherManager.activate(code, ts)	First successful LOGIN sets activation_time; later calls never move it	BR-01
VoucherManager.is_expired(code, ts)	True when ts >= activation + 24 h	BR-01
Voucher.charge_minutes(n)	Deduct n minutes, capped at the balance; returns minutes actually charged	BR-02, BR-06
Voucher.charge_mb(mb)	Round up to the next MB, deduct capped at the balance; returns MB charged	BR-03, BR-06
Voucher.is_exhausted()	True when the balance is 0	BR-02, BR-03
Voucher.devices_used	Set of MACs that ever logged in; its size is the devices_used column	BR-07
VoucherManager.report_rows(date, now)	Rows for vouchers activated on that date, sorted by voucher_code	BR-07
VoucherManager.unused_rows()	Vouchers never activated	BR-07

5. Output formats owned by the voucher module

voucher_report.csv — one row per voucher activated on a REPORT date, ordered by voucher_code:
`report_date,voucher_code,type_id,kind,activated_at,consumed,remaining,devices_used,revenue_rs,status`

consumed and remaining are minutes (integers) for TIME and MB (one decimal) for DATA. revenue_rs is the type price with two decimals. status is EXHAUSTED if the balance is 0, else EXPIRED if now >= activation + 24 h, else ACTIVE (evaluated at the last processed event time).

unused_vouchers.csv:
`voucher_code,type_id,issued_at`

6. Test plan (tests/test_vouchers.py)

Test	Checks	Rule
test_load_types_and_vouchers	4 types and 60 vouchers load; all 60 unused before any login	BR-07
test_activation_starts_24h_window	Second activate() never moves activation_time	BR-01
test_24h_boundary	23:59:59 after activation is valid; exactly 24 h is expired	BR-01
test_data_rounds_up_and_stops_at_zero	24.6 MB charges 25; balance never below 0; exhausted flag set	BR-03, BR-06
test_time_never_negative	Charging 70 min on a 60 min voucher charges 60	BR-02
test_report_revenue_on_activation_day	Voucher appears on its activation date only; exact row; unused count drops to 59	BR-07
test_exhausted_and_expired_status	Status EXHAUSTED beats EXPIRED; EXPIRED after 24 h	BR-07
test_bad_type_input_does_not_crash	Text quantity, zero devices, duplicate prefix give clean errors	BR-13

Run: `python -m unittest discover -s tests -v` — all 8 tests pass.

7. Edge cases from the brief

Edge case	Handling
LOGIN exactly 24 h after activation	Rejected EXPIRED (window is half-open, BR-01)
Two devices on a 60-minute voucher, both logged in 30 minutes	Balance is exhausted together at the 30-minute mark; both sessions cut at that second (BR-02)

Edge case	Handling
USAGE arriving after the device was idle-logged-out	Rejected NOT_LOGGED_IN; no deduction (BR-12)
LOGOUT for a device that never logged in	Error line NOT_LOGGED_IN (BR-12)
Data voucher with 0.4 MB remaining receives a 1 MB event	Deduct only what is left (charge capped), balance 0, all sessions on the voucher end at that timestamp (BR-03)

8. Decisions made where the brief is silent

#	Decision	Alternative
D1	Data is rounded up per event (BR-06 as written). Example: 24.6 MB charges 25 MB.	Deduct raw MB; the cut-off time of one sample voucher would change from 12:16 to 12:20.
D2	A repeated LOGIN of a connected MAC is logged as REJECTED / DUPLICATE_LOGIN in the session log but not written to errors.csv.	Also list it in errors.csv.
D3	Voucher status is evaluated at the last processed event time.	Evaluate at the end of each report date.
D4	For time vouchers, the remaining column on LOGIN/USAGE rows shows the committed balance (closed sessions only).	Subtract in-flight session time live.
D5	Sessions are not terminated when the 24 h window ends mid-session; only new logins are rejected.	Cut running sessions at the expiry time.